In [ ]:

```
#Name: Meenakshi Sangeeth
#Registration No: 240905630
#Section: PG
#Roll No: 34
```

In [ ]:

```
## Lab Manual
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In [15]:

```python
#1. Create a Series from a list of integers representing daily temperat
#Assign index labels as day of the week.
#a. Find and print the average (mean) temperature for the week.
#b. Identify and print the maximum and minimum temperatures and their r
#c. Display the temperatures greater than a specific value.
#d. Convert all temperatures to Fahrenheit.
#e. Print the days had temperatures above the average.

import pandas as pd

val= [30, 28, 32, 31, 27, 29, 31]
days = ['Sunday','Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday'

t = pd.Series(val, index=days)

print("Average temperature: ", t.mean())

print("\nMaximum temperature: ", t.max(), "(", t.idxmax(),")")
print("Minimum temperature: ", t.min(), "(", t.idxmin(),")")

t1 = t[t > 29]
print("\nDays with temperature over 29 degrees: \n",t1)

f = temps*1.8 +32
print("\nTempeature in Fahrenheit:\n", f)

t2 = t[t > t.mean()]
print("\nDays with temperature above mean:\n",t2)
```

```
Average temperature:  29.714285714285715

Maximum temperature:  32 ( Tuesday )
Minimum temperature:  27 ( Thursday )

Days with temperature over 29 degrees:
 Sunday       30
Tuesday      32
Wednesday    31
Saturday     31
dtype: int64

Tempeature in Fahrenheit:
 Monday       84.2
Tuesday      89.6
Wednesday    86.0
Thursday     82.4
Friday       81.5
Saturday     87.8
Sunday       88.7
dtype: float64

Days with temperature above mean:
 Sunday       30
Tuesday      32
Wednesday    31
Saturday     31
dtype: int64
```

In [18]:

```python
#2. Create a data frame with details of 10 students and columns as Roll
#a. Create a new column with total marks
#b. Find the lowest marks in Marks1
#c. Find the Highest marks in Marks2
#d. Find the average marks in Marks3
#e. Find student name with highest average
#f. Find how many students failed in Marks2 (<40)

import pandas as pd
data= {"Roll Number" : [1,2,3,4,5,6,7,8,9,10],
       "Name":['Elsa', 'Jasmin','Dheeraj','Elvin','Achal','Adil','Hiran
       "Gender":['female','female','male','male','male','male','male','
        'Marks1': [21, 26, 22, 27, 23, 28, 24, 29, 25, 30],
        'Marks2': [30, 25, 29, 24, 28, 23, 27, 22, 26, 21],
        'Marks3': [50, 45, 40, 35, 30, 32.5, 37.5, 42.5, 33, 40]
}
df = pd.DataFrame(data)

df['Total Marks'] = df['Marks1'] + df['Marks2'] + df['Marks3']
print(df)

m1 = df['Marks1'].min()
print("Lowest mark in Marks1:",min1)

m2 = df['Marks2'].max()
print("Highest mark in Marks2:",max2)

avg = df['Marks3'].mean()
print("Average mark in Marks3:",avg)

df['Average Marks'] = df[['Marks1', 'Marks2', 'Marks3']].mean(axis=1)
x = df.loc[df['Average Marks'].idxmax(), 'Name']
print("Student with highest average: ", x)

f = df[df['Marks2'] < 23].shape[0]
print("Number of students that failed in Marks2: ", f)
```

```
   Roll Number      Name  Gender  Marks1  Marks2  Marks3  Total Marks
0            1      Elsa  female      21      30    50.0        101.0
1            2    Jasmin  female      26      25    45.0         96.0
2            3   Dheeraj    male      22      29    40.0         91.0
3            4     Elvin    male      27      24    35.0         86.0
4            5     Achal    male      23      28    30.0         81.0
5            6      Adil    male      28      23    32.5         83.5
6            7     Hiran    male      24      27    37.5         88.5
7            8  Aiswarya  female      29      22    42.5         93.5
8            9   Snigdha  female      25      26    33.0         84.0
9           10     Priya  female      30      21    40.0         91.0
Lowest mark in Marks1: 35
Highest mark in Marks2: 85
Average mark in Marks3: 38.55
Student with highest average:  Elsa
Number of students that failed in Marks2:  2
```

In [ ]:

```python
## Additionals
```

In [27]:

```
#1. You are given a CSV file named "sales_data.csv" containing sales re
#a. Load the CSV file into a DataFrame.
#b. Count and display the number of rows that have at least one missing
#c. Fill the missing values using forward fill (ffill).
#d. Count and display the number of rows where the "Sales" column value

import pandas as pd
df = pd.read_csv("sales_data.csv")

c=df.isnull().sum()
print("Number of rows that have at least one missing (null) value: ",c)

c1=(df['sales']>500).sum()
print("Number of rows where the 'Sales' column value is greater than 50
```

```
Number of rows that have at least one missing (null) value:  sale_period
0
product        2
sales          1
dtype: int64
Number of rows where the 'Sales' column value is greater than 500:  9
```

In [29]:

```
#2. You are given a CSV file named "sales_data.csv" containing sales re
#Load the "sales_data.csv" file and display the first five records.
#Load the "sales_data.csv" file and display the last five records

import pandas as pd
df = pd.read_csv("sales_data.xlsx")

print("First 5 records:\n", df.head())
print("\nLast 5 records:\n", df.tail())
```

In [23]:

```
#4.You are given an Excel file named "sales_data.xlsx" containing sales
#a.Load the "sales_data.xlsx" file into a Pandas DataFrame.
#b.Use label-based indexing (.loc) to retrieve and display the record fo
#c.Use label-based indexing (.loc) to display the "Product" and "Sales"
#d.Use label-based indexing (.loc) to display records from "2024-01-02"
#e.Use label-based indexing (.loc) to select all rows where the "Product
#f.Modify the "Sales" value for "2024-01-03" to 550 using .loc, then dis

import pandas as pd

df = pd.read_excel("sales_data.xlsx")
print(df)

print("\nRecord of 2024-01-03\n", df.loc[2])

x = df.loc[4,['Product', 'Sales']]
print("\nProduct and sales of 2024-01-05\n",x)

print("\nRecords from 2024-01-02 to 2024-01-05:\n", df.loc[1:4])

print("\nRecords with Product A:\n", df.loc[df['Product'] == 'Product A

df.loc[2, 'Sales']=550
print("\nUpdated row:\n",df.loc[2])
```

```
         Date    Product  Sales
0  2024-01-01  Product A  450.0
1  2024-01-02  Product B  510.0
2  2024-01-03  Product C    NaN
3  2024-01-04  Product A  600.0
4  2024-01-05  Product B    NaN
5  2024-01-06  Product C  700.0

Record of 2024-01-03
 Date       2024-01-03
Product     Product C
Sales             NaN
Name: 2, dtype: object

Product and sales of 2024-01-05
 Product    Product B
Sales            NaN
Name: 4, dtype: object

Records from 2024-01-02 to 2024-01-05:
          Date    Product  Sales
1  2024-01-02  Product B  510.0
2  2024-01-03  Product C    NaN
3  2024-01-04  Product A  600.0
4  2024-01-05  Product B    NaN

Records with Product A:
          Date  Sales
0  2024-01-01  450.0
3  2024-01-04  600.0

Updated row:
 Date       2024-01-03
Product     Product C
Sales           550.0
Name: 2, dtype: object
```

In [ ]:

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